

Ivan Evdokimov, Ph.D.

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Professional Summary

Python Developer and Software Engineer with 6+ years of Python spanning professional roles and a PhD in computational finance and algorithmic trading, during which I developed and back-tested AI-driven trading and investment strategies using fundamental financial data sourced via Bloomberg, with published peer-reviewed output. Proficient in C++ for high-performance financial data pipelines and experienced across Python, Java, Go, SQL, and cloud-native AWS infrastructure, I have also built real-time trading simulation systems with live order entry, portfolio tracking, and PostgreSQL backends — directly analogous to OMS design. A strong collaborator, I have a consistent track record of working effectively with traders, quants, academics, and non-technical stakeholders to deliver reliable, high-quality software.

Experience

Software Engineer – GenAI, University of Essex — Colchester, UK Nov 2025 — Present

- Architected production-grade CI/CD pipelines using GitHub Actions and Vercel, enabling automated deployment with 90%+ code generation rate and weekly releases.
- Designed automated testing and QA framework for AI-generated code, cutting manual code review time by 70% while preserving production stability and security standards.
- Developed end-to-end MLOps infrastructure for a multi-tenant SaaS platform serving global B2B clients, accelerating feature delivery from monthly to weekly releases.

Software Engineer – GenAI, UK Data Service — Colchester, UK Jan 2023 — Oct 2025

- Trained, fine-tuned, and deployed domain-specific GenAI models (scikit-learn, PyTorch, Hugging Face, vLLM, Ollama) achieving 95%+ accuracy on sensitive datasets.
- Designed serverless ML infrastructure on AWS (Lambda, Step Functions, EFS) with automated retraining pipelines, reducing manual validation workload by approximately 60%.
- Optimised statistical algorithms by migrating from R to C++ using bitmask techniques, achieving 11x performance improvement enabling real-time processing of 500k+ record datasets.

Research Officer & ML Instructor, University of Essex — Colchester, UK Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime by 8x for cross-university collaborations in the UK and USA.
- Taught Data Structures, C/C++, and Machine Learning to 100+ undergraduate and postgraduate students.

Education

University of Essex, PhD in Computational Finance Oct 2021 — May 2025

- Designed and back-tested AI-driven trading and investment strategies using fundamental financial data, applying machine learning and Bayesian approaches to financial forecasting and signal generation.
- Sourced and processed market and fundamental financial data via Bloomberg, using C++ for high-performance data pipeline generation feeding AI-driven strategy research.
- Developed a PyTorch-based transfer learning framework for Bayesian model averaging, directly applied to financial return prediction and portfolio optimisation.
- Published as main author in an internationally recognised journal and co-authored 3 peer-reviewed conference papers; released open-source package adopted externally.

University of Essex, MSc in Financial Econometrics Oct 2020 — Sep 2021

- Built a C++ agent-based financial simulation modelling macroeconomic dynamics under negative interest rates, demonstrating low-latency financial systems experience.

Projects

FinBot: Chat-Based Trading Simulation github.com/ivan020/FinBot

- Built a real-time trading simulation system in Go with live order entry, a backend API recording stock transactions in PostgreSQL (Supabase), and real-time P&L tracking — directly analogous to order management system (OMS) design.
- Developed a JavaScript UI for real-time portfolio performance visualisation with interactive graphs and tables.

Chat-LLM github.com/ivan020/ChatLLM

- Built a Python-based self-hosted Ollama MCP server on Raspberry Pi with live chat integration, demonstrating low-level Python deployment and server management.

MTG Card Reader ivan020.github.io/MtgRecognizer/

- Built a Golang REST API with Mistral LLM integration and PostgreSQL backend, enabling automatic trading card recognition from photos with real-time price lookup.
- Developed a React frontend for card cataloguing, filtering, and deck management — reaching 50 monthly active users.